Espiner — Bag Strength Test Development

Bag Clamp Fixture | Design Session Progress Report

F
Suhas Sundaresh K — simple d
PERIOI
16–24 March 2026 | Mar

CONFIDENTIAL · ISO 13485:2016 · ALL MATERIALS SS 316 THROUGHOUT · FULL MATERIAL CERTS REQUIRED

A full design session was completed for the mechanical clamping fixture holding a **ripstop nylon conical bag** during destructive compression testing at **590 N**. A component geometries are defined, key design decisions are locked, and structural calculations are verified. The design is ready for Fusion 360 CAD finalisation primary measurement task remains before machining release.

ACCOMPLISHMENTS THIS SESSION

- **Bag geometry confirmed** — Clamp plane Ø67.8 mm, upper ref Ø73.2 mm, taper ~8.1° half-angle
- **Clamping principle finalised** — Conical SS surfaces + 1 mm Shore 20A silicone liners. Dual grip: threaded rod (torque-controlled) + self-energising wedge at 8.1° self-locking boundary
- **All 7 component dimensions defined** — Bottom plate, ×4 standoffs, top plate, split collar, support ring, silicone liners, closing rods
- **Structural calculations verified** — All safety factors confirmed (see table)
- **5 documents produced** — ENG-CLJIG-001 Rev E, weekly report, CAD dimensions ref, ring cross-sections, this summary
- **9 hours effort** — 5 hrs CAD + 4 hrs design development

STRUCTURAL CALCULATIONS VERIFIED

CHECK	RESULT	SF	
Standoff stiffness	k = 2,897 N/mm → δ = 0.19 mm at 560 N	—	✓
Standoff fatigue	14.1 MPa bending vs 120 MPa endurance	8.5×	✓
Ring wall (min)	3.4 mm wall, thick-walled cylinder	9.5×	✓
Friction grip	697 N grip > 590 N test load	>1×	✓
Dovetail root stub	16 mm — adequate	—	✓
Assembly clearance	2.5 mm each side — elastomer through bore	—	✓

KEY DESIGN DECISIONS LOCKED

Dovetail Retention
Through male dovetail rail (60°, 12 mm). Collar slides into plate — shoulders resist uplift at bag failure. Zero retention fasteners.

Leadscrew Closing
Half A = fixed jaw. Half B dri rods. Guided by dovetail wa skew regardless of tightenin

Half A Adjustability
M6×0.75 fine-pitch screw + slotted lock bolts. Bore centreline alignable with machine axis. Position recorded in DHF per session.

Bore Surface Finish
Ra 3.2–6.3 µm — deliberate silicone RTV mechanical ke no finishing pass — reduces

Base Plate Fixing
5-bolt pattern: 4× #10-32 UNF + 1× ½-20 UNC. Countersinks on top face. No Mark-10 modification required.

Elastomer Cone — Rejec
Cone on TP to stabilise ring Ring already stabilised by c wall. Revisit after first test d

PENDING BEFORE MACHINING RELEASE

ITEM	ACTION REQUIRED
Bag taper angle	Measure 5+ bags at 4+ heights — all dims depend on this
Top plate dims	Confirm rectangle dimensions in Fusion 360
Half A fixing holes	Confirm positions — clear of dovetail & rod holes
M8 rod hole positions	Min 12 mm clearance above/below bore edges
Standoff bolt circle	Maximise spread within plate footprint
Clamping torque	Establish from first test — suggested 5 Nm per rod
Rod & fastener lengths	Calculate from assembled CAD model

IMMEDIATE NEXT STEP

Measure 5+ bag samples at 4+ heights each to confirm taper angle. This single task unblocks all remaining bore and ring dimensions and enable machining release. Current 8.1° is derived from 2 measurements on 1 bag only — insufficient for production.